

Cover the answer · attempt aloud · check · re-test the misses tomorrow

EMBRYOLOGY — HEART TUBE, LOOPING, SEPTATION

10 CARDS

1 Which is the first organ to become functional in the human embryo, and at what week does it start beating?

The **heart**. It begins beating around **week 4**.

2 What is the embryologic consequence of an L-loop of the heart tube?

Dextrocardia — the heart sits on the right side.

3 Which septum is the last to close during atrial septation, and what defect results from its failure?

The **foramen ovale** closes last (functionally at birth; anatomic fusion over weeks). Failure = **patent foramen ovale (PFO)**.

4 Cryptogenic ischemic stroke. Bubble study shows bubbles in the left heart. Most likely mechanism?

Failed fusion between septum primum and secundum → **patent foramen ovale**; paradoxical embolism.

5 Which interventricular septum portion closes last, and where do most VSDs occur?

The **membranous** IVS closes last; most VSDs occur there.

6 Which embryologic structure, when it fails to fuse with the primum septum, produces a primum-type ASD? Classic syndrome?

The **endocardial cushions**. Classic in **Down syndrome**.

7 Embryologic mechanism of transposition of the great arteries?

Failed spiraling of the aortopulmonary septum. Aorta arises anteriorly from RV, pulmonary trunk posteriorly from LV.

8 Embryologic mechanism of persistent truncus arteriosus?

Failure of the AP septum to form/divide — a single outflow vessel persists.

9 Embryologic mechanism of tetralogy of Fallot?

Anterosuperior deviation of the infundibular septum due to abnormal neural crest cell migration. Yields VSD + overriding aorta + RV outflow obstruction + RV hypertrophy.

10 Why does DiGeorge syndrome (22q11.2) cause conotruncal defects?

Outflow tract septation requires **neural crest cells**, which are abnormal. Also explains thymic aplasia and parathyroid hypoplasia.

PHARYNGEAL ARCH DERIVATIVES

5 CARDS

11 Adult artery derived from the 3rd pharyngeal arch?

Common carotid + proximal internal carotid.

12 Ductus arteriosus is the embryologic derivative of which arch?

The **6th (left)** pharyngeal arch artery.

13 Cranial nerve paired with the 6th pharyngeal arch?

CN X (vagus) — specifically, the recurrent laryngeal branch.

14 Why does the left recurrent laryngeal nerve hook under the aortic arch?

It loops around the 6th arch derivative, which on the left becomes the ligamentum arteriosum, tethered to the arch.

15 Which arch gives rise to the proximal right subclavian artery?

The **4th** pharyngeal arch (right side).

FETAL CIRCULATION & NEONATAL FLIP

6 CARDS

16 Which fetal vessel carries the highest oxygen saturation?

The **umbilical vein** (~80% SpO₂).

17 Which large fetal vessel in the body carries the highest SpO₂ once mixed?

The **IVC** (after ductus venosus dumps oxygenated UV blood into it; ~67%).

18 Three in utero shunts and their postnatal remnants?

Ductus venosus → ligamentum venosum · foramen ovale → fossa ovalis · ductus arteriosus → ligamentum arteriosum.

19 Two physiologic changes that close the ductus arteriosus at birth?

↑ **oxygen tension** and ↓ **prostaglandin E₂** (loss of placental PGE source).

20 Drug that keeps a PDA open? Drug that closes one?

Keep open: **PGE1 (alprostadil)**. Close: **indomethacin**.

21 Three hemodynamic changes at birth?

↓ PVR (lungs inflate, ↑PaO₂); ↑ SVR (placenta removed); ↑ LV contribution to systemic flow.

22 Most anterior cardiac chamber? Clinical relevance?

Right ventricle. Penetrating chest wounds at left sternal border typically injure the RV → tamponade.

23 Most posterior cardiac chamber? Symptoms when it enlarges?

Left atrium. Compresses esophagus (dysphagia) and left recurrent laryngeal nerve (hoarseness).

24 On PA CXR, which chamber forms the right border of the cardiac silhouette?

The **right atrium**.

25 Right middle lobe pneumonia obliterates which cardiac border? Name of the sign?

The right atrial (right heart) border — the **silhouette sign**.

26 Stab wound at the left 5th ICS at the midclavicular line — what is most likely injured first?

The **left lung**. The cardiac apex is at this landmark but the lung overlies it.

27 Where do ~90% of thrombi form in atrial fibrillation, and why?

The **left atrial appendage** — pectinate-muscle stasis with poor atrial contraction.

28 Surface landmark for auscultating the tricuspid valve?

Left lower sternal border, 4th–5th ICS.

29 IV drug user, new murmur, fever, cavitary lung lesions. Valve and organism?

Tricuspid endocarditis, most commonly *Staphylococcus aureus*.

30 Define coronary dominance.

The vessel that gives rise to the **PDA** is the dominant one. ~75–80% right · ~5–10% left · ~10–20% codominant.

31 In left-dominant circulation, which artery supplies the AV node?

The **left circumflex** — whoever feeds the PDA feeds the AV node.

32 Isolated RV infarct with a normal LV — vessel? dominance?

RCA in a left-dominant patient. RCA still supplies RV; LCx feeds the inferior LV via PDA so it is spared.

33 Which papillary muscle has a single arterial supply, and from which vessel?

Posteromedial papillary muscle — supplied solely by the **PDA**.

34 Acute severe MR + pulmonary edema 3–5 days post-inferior MI. Mechanism?

Posteromedial papillary muscle rupture from single PDA-derived supply.

35 Where does the coronary sinus lie? Where does it drain?

In the **posterior atrioventricular groove**; drains into the right atrium.

36 Embryologic origin of the coronary sinus? What does a dilated CS suggest?

Left horn of the sinus venosus. Dilated CS → suspect **persistent left SVC** draining into it.

37 Trace a biventricular pacemaker lead from subclavian to the LV.

Subclavian vein → SVC → RA → CS ostium → posterior AV groove → lateral cardiac vein over LV free wall.

38 ST elevation in I and aVL — culprit vessel?

Left circumflex (lateral wall infarction).

39 ST elevation in II, III, aVF — culprit vessel?

Right coronary artery (inferior wall).

CONDUCTION SYSTEM

4 CARDS

40 Location of the SA node?

Right atrium, near the junction with the SVC — subepicardial, at the upper end of the crista terminalis.

41 Location of the AV node and the anatomic triangle that defines it?

Interatrial septum just above the septal leaflet of the tricuspid valve, near the coronary sinus ostium. **Triangle of Koch** = tricuspid annulus + tendon of Todaro + CS ostium.

42 Speed order of cardiac conduction?

Purkinje > Atria > Ventricles > AV node (*Park At Ventura, AV*). Slow AV node = atrial kick.

43 Inferior MI + Mobitz I AV block. Which artery?

The **RCA** (supplies AV node in ~90% of patients).

PROCEDURES — ANATOMY APPLIED

3 CARDS

44 PCWP approximates which intracardiac pressure, and why?

Left atrial pressure. The inflated balloon creates a static column through pulmonary capillaries → pulmonary veins → LA.

45 In transseptal LA catheterization, what structure is punctured?

The interatrial septum at the **fossa ovalis** — the thin remnant of the foramen ovale.

46 CABG: surface landmarks for the great saphenous vein?

Distal: anterior to medial malleolus. Proximal: through the saphenous opening to join the femoral vein medial to the pubic tubercle.

PERICARDIUM & EFFUSION

4 CARDS

47 Classic chest X-ray finding of a large pericardial effusion?

Enlarged "water-bottle" globular cardiac silhouette with clear lung fields.

48 Two ECG findings that together suggest large pericardial effusion?

Low-voltage QRS (fluid insulates the leads) + electrical alternans (heart swings in the fluid).

49 Most common viral causes of acute pericarditis?

Coxsackievirus B and echovirus.

50 Triad of cardiac tamponade?

Beck's triad: hypotension + JVD + muffled heart sounds. Add pulsus paradoxus >10 mmHg.

51 Branches of the aortic arch in order?

Brachiocephalic → left common carotid → left subclavian.

52 Stanford classification of aortic dissection — management for each?

Type A: ascending aorta → emergency surgery. **Type B:** distal to L subclavian → medical (IV beta-blocker, then nitroprusside).

53 Where does a Stanford type B dissection most commonly originate?

Near the origin of the **left subclavian artery** — site of highest shear stress.

54 Where does blunt traumatic aortic injury most commonly occur, and why?

At the **aortic isthmus** (just distal to L subclavian), tethered by the ligamentum arteriosum.

55 Path of an atheroembolus that causes monocular vision loss + cherry-red spot?

Internal carotid artery → ophthalmic artery → central retinal artery.

56 Describe the flow in subclavian steal syndrome.

Proximal subclavian stenosis → arm activity → **retrograde flow down the ipsilateral vertebral artery** to perfuse the arm. Brain "loses" flow → vertebrobasilar symptoms + >15 mmHg interarm SBP difference.

ABDOMINAL & PELVIC VASCULAR ANATOMY

6 CARDS

57 Vertebral level of IVC formation, and from which veins?

L4–L5, by the union of the right and left common iliac veins.

58 Why is a left-sided varicocele common but a right-sided varicocele in an adult alarming?

Left gonadal vein drains into the left renal vein, which can be compressed (nutcracker). Right gonadal drains directly into the IVC, so a right varicocele suggests **IVC obstruction** — classically renal cell carcinoma with tumor thrombus.

59 Describe nutcracker syndrome.

Compression of the left renal vein between the aorta (posterior) and SMA (anterior) → ↑ pressure in left renal and left gonadal veins → left flank pain + hematuria ± left varicocele.

60 Femoral triangle borders + NAVEL acronym?

Inguinal lig. (sup), sartorius (lat), adductor longus (med). N-A-V-E-L lateral → medial: Nerve, Artery, Vein, Empty space, Lymphatics. **Vein is immediately medial to artery.**

61 Why is a femoral artery puncture above the inguinal ligament dangerous?

Can lacerate the external iliac artery or inferior epigastric artery → bleeding into the **retroperitoneal space** (not compressible). Post-cath hypotension without external bleeding.

62 The inferior epigastric artery branches off which vessel and runs in which direction?

Off the external iliac just proximal to the inguinal ligament. Runs **superomedially** up the anterior abdominal wall.

UPPER-BODY VENOUS ANATOMY

3 CARDS

63 Two veins that unite to form the brachiocephalic vein on each side?

The subclavian vein and the internal jugular vein.

64 Unilateral right face and arm swelling but normal left side. Vessel? Classic cause?

Right brachiocephalic (innominate) vein — classically apical lung tumor (Pancoast) or central-line thrombus.

65 Embryologic origin of the SVC?

The **common cardinal veins**.